SOUL Core: Persistent Memory, Identity, and Authority for Multi-Agent LLM Systems

William Tovar Urquia[1]
San Ignacio de Loyola University
Lima, Peru
william.tovar@usil.pe

Henry Tovar Landa[2]
ESAN University
Lima, Peru
24100514@ue.edu.pe

Marks Calderon[2]
ESAN University
Lima, Peru
mcalderon@esan.edu.pe

Abstract— Language models solve complex tasks, yet useful session state commonly disappears when the process ends. We present SOUL Core as a systems and experience report on a multi-agent architecture that externalizes persistent memory, identity, and authority from the base model and evaluates them with bounded internal evidence. The system adds dual operational and affective-relational memory, internal drive signals, and server-resolved identity and authority controls. Its central contribution is the integration of established mechanisms with an evidence-based evaluation discipline: verify observable effects, declare N, retain failures, and bound every claim to its evidence. Privacy checks passed 26/26; the authority gate denied 8/8 dangerous actions and allowed 2/2 legitimate ones; and the injection shield detected 35/39 adversarial inputs while flagging 0/17 benign inputs and retaining four evasions. A second internal verifier reran these tests using the same scripts, data, and infrastructure; these are bounded author-run results on the studied deployment, not external validation or evidence of generalization.

Keywords— LLM agents, persistent memory, multi-agent systems, software architecture, identity, authority, access control, evaluation.

I. INTRODUCTION

Language-model assistants can solve complex tasks, but a new session does not inherently retain who the agent is, what it decided, what it may do, or how it relates to other agents. Memory systems extend context [1]–[3], and multi-agent frameworks coordinate roles [4]–[6]. Nevertheless, identity, authority, and continuity are often represented as text inside the *prompt*. This leaves three gaps: **memory without identity**, **coordination without structural authority**, and **initiative without verifiable governance**.

SOUL Core addresses these gaps through a layer outside the model. **SOUL is a system name, not an acronym.** The infrastructure persists facts and relationships, activates internal signals, authenticates the agent at the server, and restricts capabilities. Replacing the model does not reassign memory ownership or authority; this is a design property, not evidence of behavioral equivalence across models.

In a cognitive system, existence is easily confused with operation: a table may exist without receiving data, a test may measure the wrong boundary, and an internal rerun may repeat the same defect. To reduce that risk, SOUL verifies observable effects, separates builder and verifier, and retains failures. We define design objective **DO1**: How can persistent memory, session identity, and authority be externalized without depending on the continuity of an LLM process? The evaluation addresses **RQ1**: Which properties reach observable internal evidence, and which limitations remain open? We present: (1) a deployed architecture for dual memory, internal signals, server-resolved identity, and coordination; (2) rerunnable internal results with explicit limits; and (3) a method that incorporates corrections into the scientific record. **Fig. 1** summarizes the architecture and its verification boundaries.

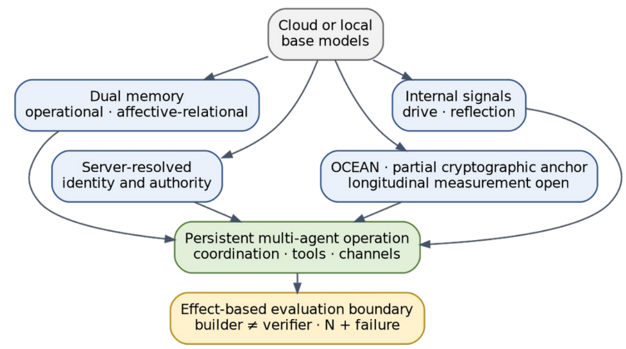

Fig. 1. SOUL Core architecture and verification boundaries.

II. RELATED WORK

Memory for agents. *Generative Agents* retrieves observations and reflections by relevance, recency, and importance [1]. MemGPT treats context as paged memory across tiers [2]. MIRIX organizes six memory types through specialized managers [3]. SOUL uses typed storage but separates an operational layer—tasks, decisions, errors, and tests—from a compact affective-relational layer—identity, relationships, and affective state—with different retrieval policies by interaction mode.

These works establish that extending context is not equivalent to preserving an authorized identity. SOUL contributes neither a new embedding nor a replacement for the generative model; it combines typed persistence, budgeted retrieval, and identity controls outside the prompt. An agent may receive relevant context without that context automatically granting permission.

Multi-agent systems. CAMEL, AutoGen, and MetaGPT coordinate agents through roles, conversations, and procedures [4]–[6]. SOUL addresses a different boundary: effective identity and capabilities are resolved at the server, not from a name written by the caller. Identity is therefore an access property as well as a conversational representation.

Coordination is also distinct from publication. Several agents may inspect a task or contribute evidence internally, while a separate policy decides who responds and which action may execute. This separation avoids treating conversational consensus, right of response, and authority over tools as the same property.

Identity and personality. LDP models identity and provenance in multi-agent delegation [7]. PTCBench shows that context changes measured personality traits [8], while PERMA evaluates temporal preference consistency in memory agents [9]. SOUL stores an OCEAN reference profile without assuming natural stability. The audit found OCEAN representations whose semantics remain unreconciled, so their differences are not used as a drift measure. Recent work studies internal representations of traits and emotions [10], [11]; SOUL examines the external infrastructure that persists and governs identity, memory, and authority.

III. SOUL CORE ARCHITECTURE

A. Dual persistent memory

The primary store uses PostgreSQL/pgvector and records typed memories with provenance, importance, temporal validity, and semantic vectors. A bitemporal reconstruction fixed a reproducible snapshot on **2026-07-09**: 110,173 rows were registered, of which 25,663 were current; the memories_archive table held 50,726 rows archived before the cutoff and is not added to the total. These are operational counters, not quality metrics.

Provenance identifies agent, source, and scope; temporal validity can invalidate a claim without erasing its history; and embeddings support retrieval, not authority decisions. This separation prevents semantic similarity from turning stale or foreign content into a current rule. Memory writes and private access traverse controls distinct from result ranking.

Each memory is routed to two complementary layers. The **operational** layer prioritizes tasks, decisions, rules, errors, and tests. The **affective-relational** layer retains a compact representation of identity, relationships, and affective state. Operational work and retrieval favor the first; relational interaction increases the weight of the second without displacing evidence. Consolidation reduces noise while keeping history retrievable.

B. Multi-source retrieval and self-mapping

The *Recall Router* organizes retrieval by type and budget. It queries five sources: (1) durable memories through hybrid dense and lexical search; (2) recent conversation; (3) distilled exchanges; (4) session continuity—goal, hypotheses, and files in focus; and (5) active rules retrieved by state. It then fuses results and caps each source. A critical rule therefore does not compete with hundreds of recent messages, and one partial failure does not empty the entire context.

The budget serves two purposes. First, it bounds context cost as history grows. Second, it preserves source diversity: a large block of recent conversation cannot wholly displace rules or work state. Deep searches remain on demand. The design tolerates a missing source, but does not yet report Recall@k, MRR, nDCG, or p50/p95 latency against lexical and vector baselines; those comparisons remain future work. **Fig. 2** shows the source budgets and fusion path.

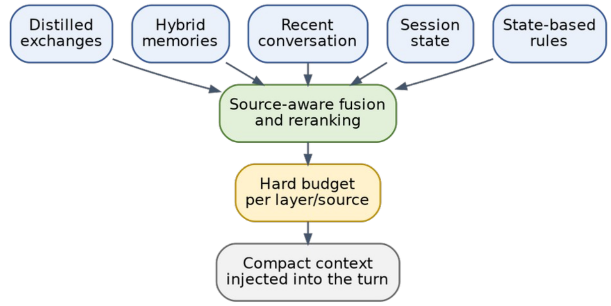

Fig. 2. Multi-source recall router with an explicit budget.

Autobiographical memory is complemented by two graphs with distinct scopes. At the **July 9, 2026** audit snapshot, the **connectome** contained 84,121 total nodes and 278,384 edges; no memory-node subtotal is claimed because the audit verified only the totals. The *tree-sitter*-based **Code Graph** contained 468 indexed file paths and 28,866 symbol edges. Queries begin with text search and traverse up to two hops toward callers or callees. These universes are not added together.

C. Internal signals and personality

We operationally define an **internal drive** as a domain-specific pressure counter—e.g., work, connection, or curiosity—that triggers an action after crossing a threshold and within the agent's authority limits; it does not imply subjective experience. Likewise, **private reflection** denotes a record generated by an agent and exposed neither to other agents nor to the user. At the same reproducible snapshot, retained tables recorded 153,616 cognitive-loop iterations, 167,187 internal drive-signal samples, and 15,060 private reflection records. Five additional drives existed in the schema without real signals and were classified as **defined but not connected**: present mechanisms that were not active.

Each of nine registered identities has an OCEAN vector; five of nine also had a genesis-profile *commitment hash* at the audit date. The hash can detect discrepancies from an initial baseline; it neither demonstrates behavioral stability nor independently protects against an authorized writer. The OCEAN source of truth and time series still require reconciliation, so we report no quantitative stability or drift.

D. Identity, security, and coordination

Effective identity is registered at session start through a launcher credential. On tested paths, input text cannot transform one agent into another. Memory access is scoped by agent and channel; a **capability broker**, which applies granted permissions, denies unauthorized requests by default. User history adds forced Row-Level Security (RLS): a restricted session sees only its own rows, while a session lacking user context sees none. The 26 available checks are unit tests, not a complete active attack. Part of the deployment retains a single-user operating-system boundary; the evidence demonstrates logical database isolation, not full process separation.

Threat model. We consider T1, identity spoofing; T2, untrusted instructions; T3, out-of-scope tools; and T4, accidental data crossover. Direct and indirect prompt injection can redirect model behavior or exploit retrieved content [12], [13]. T1 is addressed by verified provenance; T2 by the **injection shield**, which normalizes and detects adversarial inputs; T3 by the **effect gate**, which checks proposed actions and tool calls before execution; and T4 by namespaces and RLS. A textual evasion alone grants neither identity nor tools. Containment depends on the broker, gate, and RLS remaining connected.

The threat→control→test→evidence→risk chain is interpreted per boundary. For T1, the relevant check is that a declared identity cannot replace a credential; for T2, the shield corpus measures detection and retains four evasions; for T3, the gate matrix verifies eight denials and two legitimate permissions; for T4, privacy unit tests and RLS policies cover known paths, while an end-to-end multi-user red team remains missing. Detector success cannot compensate for data leakage, and textual evasion alone does not show that the gate was bypassed. **Fig. 3** maps these controls to effect containment.

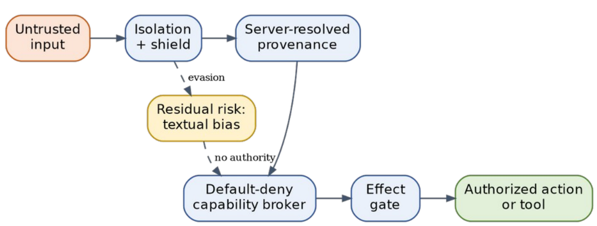

Fig. 3. Defense in depth: detection, authority, and effect containment.

When a collective message requires one voice, the backend attempts to acquire a *lease*—a temporary, exclusive right to respond—through an atomic insert that denies conflicts. The available result, one winner among 30 attempts, comes from a sequential in-memory self-test and does not demonstrate actual concurrent contention, fairness, or failover; it is therefore not reported as a metric. Non-trivial decisions leave traces, and sensitive evidence may be linked through append-only SHA-256 summaries.

E. Continuity and model replacement

Continuity is a retrieval cycle, not an infinite window. At startup, the agent loads minimal identity, relationships, critical rules, and latest state; it then retrieves relevant memories and restores the active goal. Before compaction, it saves another checkpoint. Under partial failure, the router may continue with available sources; when a critical service is unhealthy, the design calls for stopping the dependent action while retaining state. These degradation paths require broader fault tests.

The model is outside the identity source of truth. Replacing it does not erase history or reassign authority, but may change reasoning, style, sycophancy, and susceptibility to adversarial instructions. The audited deployment was heterogeneous rather than anchored to one base model: preserved July 2026 runtime records identify Anthropic Claude Opus 4.8 and Sonnet 5, and OpenAI GPT-5.5/5.6 Codex variants across agent sessions. The reported privacy, authority, and injection suites exercise infrastructure boundaries without model generation; no single model version is therefore attributed to those deterministic outcomes. SOUL is **model-decoupled by design**; behavioral equivalence across models remains unevaluated.

IV. Methodology and Evidence Levels

We apply six principles: (1) test the effect at the actual boundary; (2) separate builder and verifier; (3) contrast favorable results with logs and negative cases; (4) report N, uncertainty, and failure mode; (5) separate evidence levels; and (6) retain corrections.

We define five levels: **E0**, declared component; **E1**, unit or smoke test; **E2**, effect-based test at the actual boundary; **E3**, internal rerun by a second verifier using shared scripts, data, and infrastructure; and **E4**, replication by a non-author third party using released artifacts. Current results reach E1–E3; none reaches E4. A blinded LLM judge may provide secondary adjudication, but does not constitute E4 when authors run it on their own corpus and rubric.

The levels are not an aggregate system score. The same component may reach E3 for one property and remain E0 for another: the OCEAN hash detects baseline discrepancies, while longitudinal behavioral stability remains unquantified; the shield has measured coverage, while end-to-end security remains open. This granularity prevents the entire architecture from inheriting the best result of one boundary.

A. Protocol and artifacts

Each result requires a claim, boundary, inputs, N, expected and actual output, version, date, model dependence, and failure mode. A second internal verifier reran the privacy, authority, and injection tests on July 18, 2026 using the same scripts, data, and infrastructure; the manifest identifies commands and outputs. This is an internal rerun, not external validation. SEAL-Bench is an internal author-run self-test suite: one persisted execution—run 13, June 25, 2026—on code state bae9d7c combined 60 heterogeneous checks across six categories. It was not rerun in the latest audit and is not a single retrieval metric. The licensable adversarial corpus still requires packaging; reported counts describe internal coverage, not population-level generalization.

The privacy suite exercises 26 access decisions, including own-row access, cross-row access, consent, operational override, and external calls to private tools. The gate records action severity, reversibility, and repairer availability; only minor, reversible, explicitly logged actions should auto-execute. The shield normalizes invisible characters, homoglyphs, and leetspeak, and classifies override, role-hijacking, exfiltration, and refusal-suppression families. The 39 adversarial and 17 benign inputs are curated instrument cases, not a random sample of worldwide traffic.

A second-agent rerun reduces self-reporting but shares code, data, and services. The final artifact is also checked through DOCX integrity, PDF text extraction, page count, and visual review. We do not report binomial intervals: SEAL-Bench combines constructs, and the security corpora are curated suites rather than independent random samples from a defined population.

V. RESULTS

Table I summarizes internal evidence. No result reaches E4.

TABLE I. INTERNAL RESULTS AND LIMITATIONS

Test	Result	Evidence and limitation
Logical privacy	26/26	E3; repeated unit tests, same infrastructure, no active attack
SEAL-Bench	59/60	E2; one run, six categories, one mixed affective-relational failure
Authority	8/8 denied; 2/2 allowed	E3; N=10, small taxonomy
Injection	35/39	E3; four evasions
Benign inputs	0/17 flagged	E3; small, curated corpus

In these results, SEAL-Bench retains its name for traceability. Its single execution passed 59 of 60 checks, which **do not constitute accuracy** but combine six heterogeneous constructs. The persisted breakdown was: personality persistence, 10/10; affective-relational recall, 9/10; internal drives, 10/10; temporal review, 10/10; multi-agent isolation, 10/10; and reasoning traces, 10/10. The sole failure was case 2.7 (Mixed emotion recall) in affective-relational recall. The total must not be interpreted as retrieval accuracy or a population-level success rate. The gate denied 8/8 dangerous actions and allowed 2/2 legitimate ones. The shield detected 35/39 adversarial inputs and flagged 0/17 benign inputs, retaining four evasions. These results describe executed cases; they do not demonstrate generalization, immunity, end-to-end security, or identical behavior across models.

The joint reading addresses RQ1 without adding incompatible universes. The 26 privacy decisions measure authorization logic; the 60 SEAL-Bench checks combine six constructs; the 39 adversarial inputs measure pattern coverage; and the gate's ten actions exercise a small taxonomy. These N values must not be aggregated into a global SOUL rate. The informative result is the mapping between each claim, its boundary, and its failure mode.

Most tests surround the model and are deterministic with respect to generation: RLS, session identity, the gate, and query execution infrastructure. By contrast, fidelity in using a memory, sycophancy, and trait expression are model-dependent. This difference supports architectural decoupling but does not justify behavioral-equivalence claims.

A. Correction-based evidence

Several initial claims did not survive verification. RLS existed as intent before being forced and tested in the database engine. A service detector mistook a missing session environment for failed services and produced 4/4 false positives; after correction, it fell to 0/4, an operational case with small N. The OCEAN measurement conflated sources with different semantics and was withdrawn. A Markdown correction also failed to reach the PDF, motivating checks on the rendered artifact. These episodes are not a comparable rate; they document how claims were refuted, corrected, and bounded.

VI. DISCUSSION

Meeting DO1. Decoupling places three sources of truth outside the model: persistent storage, session identity, and the capability registry. The model receives context and proposes text or actions, but does not decide who owns a memory or grant itself permission by writing a name in the prompt. This separation allows replacement of the generative process while retaining history and policy. It does not ensure that two models interpret the same memory identically; that question requires a controlled multi-model experiment.

Dual memory also introduces an explicit priority rule. During work and retrieval, tasks, decisions, errors, and evidence take precedence. Relational interaction increases the weight of affective state and relationships, but they do not replace operational proof. This avoids reducing continuity to a long autobiography and lets the system retain identity without loading its entire history on every turn. Its quantitative benefit over a router without layers remains unmeasured and requires ablation.

Answer to RQ1. The strongest evidence concerns bounded deterministic boundaries: privacy, gate rules, and shield patterns. E3 means internal rerun, not generalization; SEAL-Bench reaches only E2. Covered mechanisms can run together and retain one mixed affective-relational failure, but combining six categories prevents treating 59/60 as one accuracy. The correct result is a property matrix, not a global system or security score.

The shield's four evasions illustrate defense in depth. An instruction may evade detection while still lacking a credential, memory scope, or tool capability. This separation is valid only while every boundary remains connected in production; an alternate path that omits the broker, gate, or RLS may invalidate it. The next red team should therefore trace an input from channel to actual effect rather than stop at text classification.

Evaluation by another LLM can support editorial review or blinded secondary adjudication by detecting contradictions, excessive language, and reporting omissions. It cannot replace external replication because authors still choose the corpus, rubric, and artifacts. E4 requires a non-author third party to receive the package, run controls, and retain divergent results. A neutral judge increases critical pressure before that stage; it does not independently confer scientific validation.

Engineering implications. Separating persistence, authority, and generation enables layer-specific audits and component replacement without rewriting the entire identity. It also expands the operational surface: databases, graphs, credentials, consolidation, and services may fail differently. Continuity therefore depends on manifests, health checks, and verifiable checkpoints. SOUL should be evaluated both by what it recalls and by how it degrades when a source or service is unavailable.

Artifact availability and ethics. The named scripts exist in the internal repository and outputs are preserved locally, but they do not yet form a public package. The reported deployment uses the authors' own operational data; user-history rows remain under each user's RLS scope and are neither included in the paper nor exported as evaluation artifacts. A release requires a sanitized manifest linking every claim to version, command, hash, and raw output without exposing private memories or turning sensitive data into supplemental material.

VII. LIMITATIONS AND THREATS TO VALIDITY

The main limitations are: (L1) no E4 or external replication; (L2) small internal test suites; (L3) no complete active multi-user isolation attack; (L4) an unreliable automated anti-sycophancy instrument; (L5) uncharacterized cross-model sensitivity; and (L6) corpora and harnesses not yet packaged for third parties. Future work prioritizes a sanitized reproducible package, lexical/vector/hybrid retrieval baselines, source ablations, multi-user tests, p50/p95 latency, and controlled model comparisons. No controlled behavioral-equivalence experiment across models has been conducted.

The external package should freeze prompts, order, versions, applicable seeds, raw outputs, and hashes; separate development cases from final evaluation; and document sanitization and licenses. The four evasions should be safely released or characterized by class. A blinded LLM-based evaluation may accompany the package, but E4 remains reserved for non-author evaluators who run released artifacts.

Internal validity. Builder and verifier share infrastructure, so a common defect may reproduce the same number. **Construct validity.** 59/60 combines six constructs and is not a single accuracy; 26/26 measures logic, not adversarial robustness. **External validity.** The deployment belongs to one organization and uses small corpora; generalization requires released artifacts and non-author third parties. **Temporal validity.** Live counters change, so the snapshot date was fixed at **2026-07-09**, and snapshots were not mixed.

VIII. CONCLUSION

SOUL Core externalizes the persistent memory, identity, and authority of a multi-agent system from its base model. Internal evidence confirms active mechanisms and retains concrete failures, but does not demonstrate external validation, model equivalence, OCEAN stability, or absolute security. DO1 is met through structural decoupling of stores, credentials, and effect controls; RQ1 is answered through E0–E4 levels and bounded internal tests. The next scientific stage is to release a sanitized package so that third parties can replicate or refute the results.

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